

User Story Splitting in Agile Software Development using Machine Learning Approach

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Abstract— User story forms the requirements for agile software development in textual format. The user story should be in the proper template so that developers can implement it easily. The quality of the user story should also be maintained semantically. As semantics gives meaningful artifacts about the customer requirements. User stories can be written easily, if there is complexity then need attention to make it easy. As user story splitting techniques are available to split various user stories. In this paper, we proposed a user story-splitting technique in agile software development using a machine learning approach. Data entry and CRUD-based user story-splitting techniques are used to split the complex user story. Also, user story distribution uses some machine learning approaches such as K-medoids and K-means clustering based on similarity measures. To measure the performance of the approach, silhouette value is calculated which shows the cohesion between user stories. As result shows that K-means gives the value 0.114 for cluster size 15 whereas, K-medoids gives 0.068 value. K-means is far better than K-medoids clustering. This approach reduces the software development time of requirements and improves performance as well.

Keywords— *K-medoids, K-means, Splitting, Silhouette Value, Similarity measure.*

I. INTRODUCTION

Agile methodology improves the performance of the organization through customer interaction during the development of a large system[1]. As developer and customer relationship is the key feature to developing the software better with high-quality features. Agile is capable to develop the project iteratively and incrementally, where coding and testing have done together. But before proceeding to the coding section, the initial requirement phase is done positively so that coding can be implemented easily. Requirement engineering[2] is phase of agile software development (ASD), in which requirements should be written natural language. This is also known as textual requirements or user story in case agile scrum. Mostly practitioners adopted the user story to write the requirements in templates which are given by many authors. User story adopted by developers approximate 70% in ASD[3]. But template given by Mike Cohn[4] highly recommendation by practitioners. In this template, represents the three sections: who, what and who for example As a “who” I want “what” so that “why”.

Day by day the size of software and customer involvement increased in the development process same and

the size of the requirements also increased. Complete software system breaks into modules to handle the requirements separately or module-wise development. The definition of user stories should also be maintained in a way that development teams can implement easily. These user stories are maintained in a manual way which is a very time-consuming process[5], need an automated tool to convert the whole requirements into a user stories template through natural language processing. The breakdown process of large systems helped developers to understand the requirements of each module.

Splitting complex user story into small tasks is common[6], while working with user stories. But user story splitting is a very challenging task. To decide which user story needs to be divided or which one does not and how. Because if a user story is not written in a proper way it leads to confusing for the developer when going to implement it. A story has a scope as per size if it is the large then higher scope and if is small then smaller scope. Complexity of user story depends on the scope if it is small scope then less otherwise high complex. The result of complex user story implementation affects development time and also missing the deadlines of the project [7]. One of the most popular techniques for decomposing user stories is user story mapping[8].

After user story splitting, the automated user story tool has defined the clusters of similar requirements through some clustering algorithm. K-means and K-medoids are the two clustering algorithms[9] that enable the clustering of user stories. Some authors implemented clustering algorithms on user stories using similarity measures [10][11][12][13]. But still, some lacking in the procedure of user story clustering. In this approach, the cosine similarity measure[14] is used to form clusters, which is helpful to plot the data points in a 2D environment. As machine learning improves the accuracy of various types of applications [15][16] and for agile user story it is also beneficial.

The contribution of our work is to break the complex user story into small user stories so that implementation can be done easily, identification and understanding requirements, and reduce the development time of agile software.

Sections of this paper are as follows: Sections II and III represent the related work and user story-splitting techniques explained. The proposed approach and user story quality are both explained in Section IV. Section V result discussion and performance analysis done in this. Finally, the last section concludes the work.

II. RELATED WORK

Garm Lucassen et al. [3] proposed framework named quality in user story (QUS) which has 13 quality criteria. Under this framework, have the automatic quality in user story artisan that detects quality and also suggest possible solutions. It also splits the user story when a single user story has two different requirements under the atomic phase. User story also splits based on role. The framework was implemented on 1023 user stories from 18 case studies. It automated 100% recall and high enough precision. Davide Taibi et al. [17] gives a comparison of requirements splitting tasks of the unstructured process. The authors conducted a multiple-case study with four teams. As result shows that teams split the requirements by using various techniques based on the selected development process. All these techniques were implemented on agile scrum, scrum with Kanban, XP (eXtreme Programming), and banana development. The most effective output given by scrum with Kanban and XP. Angela Cristina Pereira [18] suggested a model which generates the user stories using NLP concept. Author considered 3 groups and group third gives the best result than other two. In future aspect this paper gives the direction to add the lemmatization to identify the system entity of their core word. Muhammad Bilal Shahid [7] proposed supervised machine learning based user story splitting technique. This work used 4 types of learning techniques listed as random forest, support vector machine, K-NN and decision tree. These learning techniques implemented on 3 open-source software from Jira. To balance the dataset used ROSE and SMOTE. Best result is given by random forest with SMOTE which has accuracy 94.21%. In future, recommendation of tool to split user story-based developer perspective. Takwa Kochbati et al. [13] proposed a model based on machine learning which splits system in small sub-system. The type of clustering approach used is hierarchical agglomerative to form similar requirements. I. K. Raharjana et al. [19] gives literature review on the usage of NLP in user stories. This survey mentioned the template and format to be given to form the user requirements. Through software how textual content can be converted into template of user stories.

III. USER STORY SPLITTING

In ARE, writing user story is not an easy task as they can get pretty big if no one points out[20]. In INVEST rule S stands for small and E stands for an estimate. Both are related to each other as a small user story can be estimated easily which leads to the ability to correctly estimate[3] A story can finish in 2 and 3 days as per estimation if not then split it. Another aspect a user story is not small enough if it cannot be tested easily as T stands for a test. So, there are various ways to split the user story [17] which are follows: -

- By roles
- By workflows
- By datatypes
- By data entry
- By complexity
- By browser/platform
- by business rules
- By CRUD type of operations

- By acceptance criteria
- Mix them all

Above listed user story splitting techniques are used to break down the user story into sub-user stories. So, the complexity of the user story can be reduced and a large user story splits into smaller user stories. Splitting is done based on “I want” or means section.

IV. PROPOSED WORK

In our proposed, user story splitting is done by data entry and CRUD technique. Both techniques are considered to break down the complex user story. In the data entry technique, consider any parameter inserted by the user whereas, in CRUD basis on user operations like create, read, update and delete.

A. User Story Quality

The quality of the user story depends on the syntactic, semantic and pragmatic [3]. As syntactic is related to ell-formed, semantic is related to meaning and pragmatic is concerned with the full sentence. Our work is finding out the quality of the user story through a semantic approach. For the semantic approach, data entry and CRUD are both techniques considered. CRUD can split the user story based on an operation like add, update, edit, delete and rename, and many more. All the operations are separated by commas and “and” and “or” may be any type of word.

As per the user story template, the “I want” section, or named as a meaning section of the user story plays a role in to split. After applying data entry and CRUD techniques as per demand, the user story splits into more user stories. When this means the section of the user story is separated by commas or words then need for splitting. For example, if the user story has a means section as “I want to edit and delete help wanted ads”. In this operation edit and delete two operations specified in the user story. It increases the complexity of the user story, which later splits into individual operations. Further, applying the CRUD technique, the user story can be split into “I want to edit help wanted ads” and “I want to delete help wanted ads”. User story splitting increases the simplicity of the implementation of user stories.

V. RESULT DISCUSSION AND PERFORMANCE ANALYSIS

Python is used to implements the coding section of proposed work. The dataset considered as per template [21]. The dataset text file has 97 user stories, and the ID of user story is represented as US1 to US97. Where some user stories are simple are some are complex. As complex user stories have the large meaning section. If break the meaning section into small one then it becomes very easier to implement the user stories. After go through with dataset. Simulation tool initially find out the meaning section which has multiple operation and separated by commas. Later, to perform the splitting on user stories, both techniques, data entry and CRUD implemented. After screening of all user stories find out that total six are complex, which are US47, US49, US55, US74, US76 and US86. Each selected US55 splits into small one shown in Fig 1 using data entry. As US55 is divided into US55-0, US55-1 and US55-2 user stories.

Original User Story: - As a site editor, I want to set start publishing dates, old article date, and stop publishing dates for articles, so that articles appear only during appropriate periods.
 Extraction of meaning section: - I want to set start publishing dates, old article date, and stop publishing dates for articles
 User Story 0: - I want to set start publishing dates
 User Story 1: - old article date
 User Story 2: - and stop publishing dates for article

Fig 1. User story splitting by data entry splitting technique

Whereas, CRUD splitting technique shown in Fig 2 for US47. As US47 is divided into US47-0 and US47-1 user stories. Same as US 49 split into US47-0 and US47-1, US74 is split into US74-0 and US74-1, US76 is split into US76-0 and US76-1 and US86 split into US86-0 and US86-1. Later on, meaning section written in proper format in user story template. Each split user story preceded by "I want" section.

Original User Story: - As a site admin, I want to edit and delete help wanted ads, so that I can correct small problems or make sure each ad complies with site guidelines.
 Extraction of meaning section: - I want to edit and delete help wanted ads
 User Story 0: - I want to edit
 User Story 1: - and delete help wanted ad

Fig 2. Silhouette coefficient values

TABLE I. USER STORY SPLITTING

User Story ID	Original user story [19]	User story splitting	Splitting Technique
US47	"As a site admin, I want to edit and delete help wanted ads, so that I can correct small problems or make sure each ad complies with site guidelines."	I want to edit help wanted ads	CRUD
		I want to delete help wanted ads	
		I want to be able to extend an ad for another 30 days by updating the posting	
US55	"As a site editor, I want to set start publishing dates, old article date, and stop publishing dates for articles, so that articles appear only during appropriate periods."	I want to set start publishing dates for articles	Data entry
		I want to set old article date for articles	
		I want to set stop publishing dates for articles	
		I want to activate my membership once my name has been loaded to the registry.	

All six selected user stories also divided into small user stories as shown in Table I on the basis of data entry and CRUD technique. Now after splitting 97 user stories text file converted into 104 user stories.

Added new user stories are 7 in existing dataset to make simplicity of requirements.

A. User Story Distribution

As per the performance analysis, all user stories are distributed by K-medoids and K-means clustering algorithm which is implemented based on similarity measure. In this implementation, the cosine distance similarity measure is considered to plot two-dimensional data for giving the graphical representation of user stories.

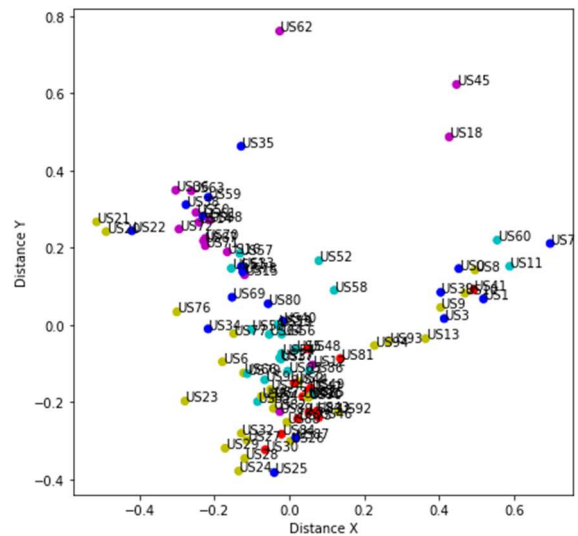


Fig 3. K-medoids based non-user story splitting

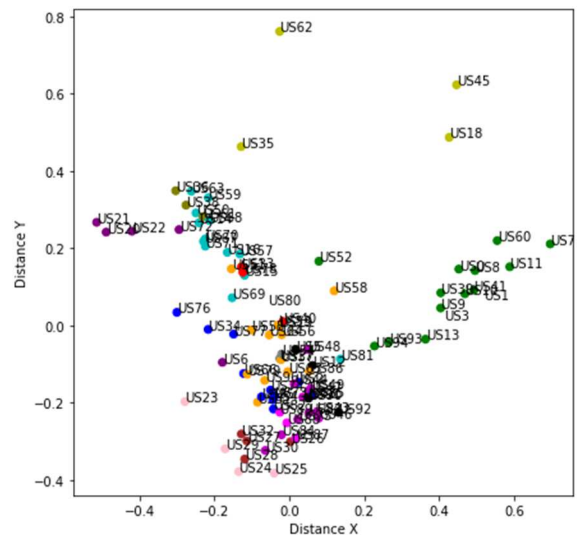


Fig 4. K-means based non-user story splitting

As Fig 3 shows the distribution of non-splitting user story using K-medoids. Whereas, the K-means-based distribution of non-splitting user story is depicted in Fig 4. In both figures, cluster size is 15 for K-medoids and K-means clustering algorithm. As in the case of a non-splitting user story, only 97 data are considered for plotting.

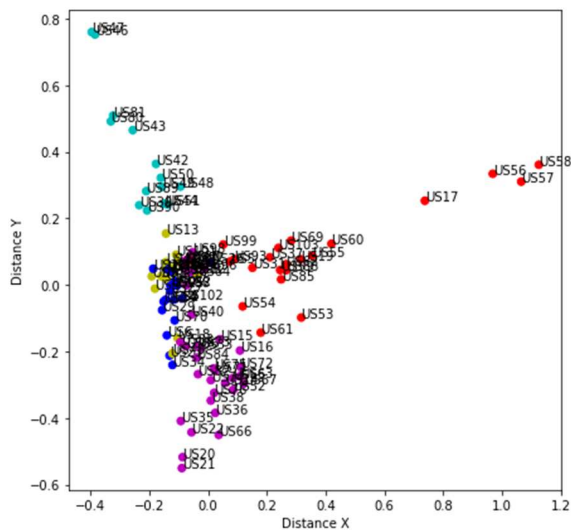


Fig 5. K-medoids based user story splitting

When splitting is done then the size of user stories is increased by 7 more stories are added to the dataset. Now Fig 5 and 6 represented the 104 user stories distribution using K-medoids and K-means clustering algorithm with the same size of the cluster which is 15.

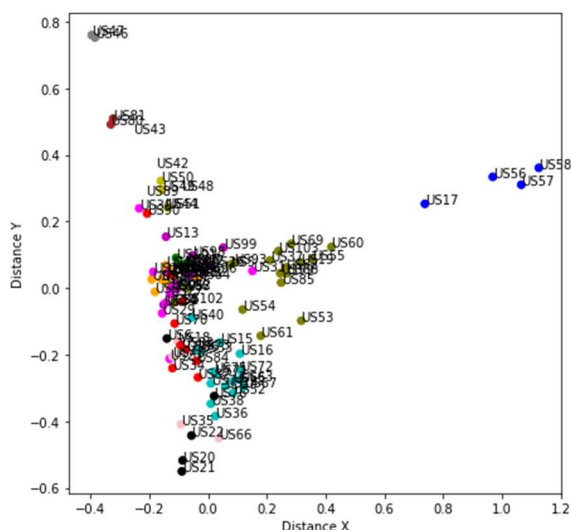


Fig 6. K-means based user story splitting

After user story distribution, find out whether the cluster formation is done accurately or not, and calculate the silhouette coefficient which tells how well the user story is placed in the right cluster. Silhouette value is in between -1 and +1, where -1 is for the worst value indication and +1 is for the best value indication. As Barbosa [10] gives K-medoids based clustering on user story distribution on similarity measures, worked on 50 user stories with an average generated cluster size is 13. In our proposed work, 97 user stories were considered, and cluster sizes start from 2 to 15. In the pre-processing phase, lemmatization is added which is not considered previously. Our work shows the comparison between K-medoids and the K-means clustering algorithm with the addition of splitting techniques.

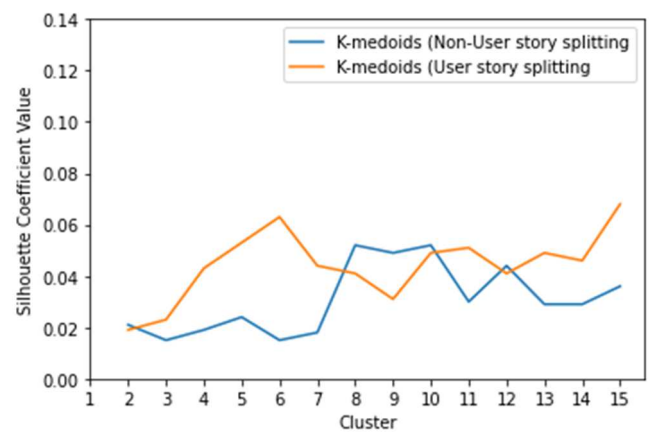


Fig 7. Silhouette coefficient values using K-medoids

Fig 7 shows the comparison of silhouette value between non-splitting and splitting with distribution through K-medoids clustering.

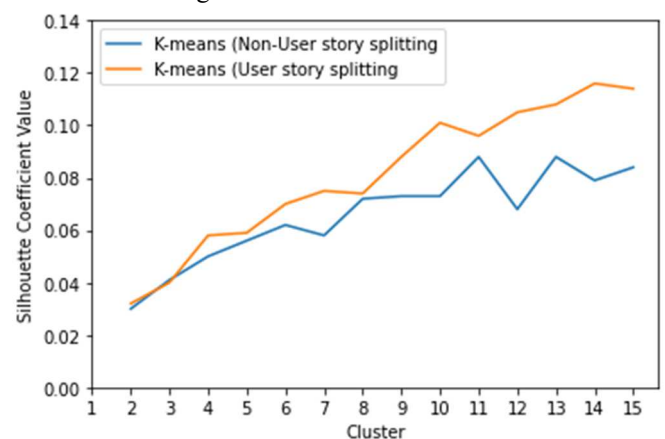


Fig. 8 Silhouette coefficient values using K-means

A comparison of silhouette value between non-splitting and splitting with distribution through K-means clustering is shown in Fig 8. And can see the difference between values and how user stories are cohesive within the cluster. As in the case of splitting, the value is increased when the cluster size is increased. In both clustering algorithms, K-means gives a more remarkable output than K-medoids. As the silhouette value using K-means with a splitting approach is high than K-medoids.

If the cohesion between user stories is good then implementation becomes also easy for developers. The understanding user story is easy for developers then they take less time to implement and also give the project on time other project development leads to failure due to time-bound. This approach not only reduces the time of development taken by developers and also gives the direction of better identification of requirements in the form of user stories.

VI. CONCLUSION

In agile software development, requirements are represented through a textual format which is the user story. User story follows the template like “who”, “what” and “why” sections. The meaning section is “what” which represents the quality of the user story. The semantics of the user story should be mentioned. If the meaning section is large then it increases the complexity of the user story. To

reduce the complexity of the user story, need to implement the splitting techniques. As two techniques we have implemented data entry and CRUD on 97 user stories to split the complex user story. Later on, clustering machine learning approaches were used for the distribution of user stories on the behalf of the similarity measure. To measure the performance of the approach, silhouette value is calculated which shows the cohesion between user stories. As result shows that K-means gives the value 0.114 for cluster size 15 whereas, K-medoids gives 0.068 value. K-means is far better than K-medoids clustering. This approach reduces the software development time of requirements and improves performance as well.

In the future, other splitting techniques such as by roles, data types, business rules, acceptance criteria, and miscellaneous can also be implemented to split the user stories. More clustering approaches can also be used for distribution and other similarity measures can also be considered.

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